



Independent Research Testing Center  
875 Innovation Drive, Cambridge, MA 02139, USA

## CERTIFICATE OF ANALYSIS (COA)

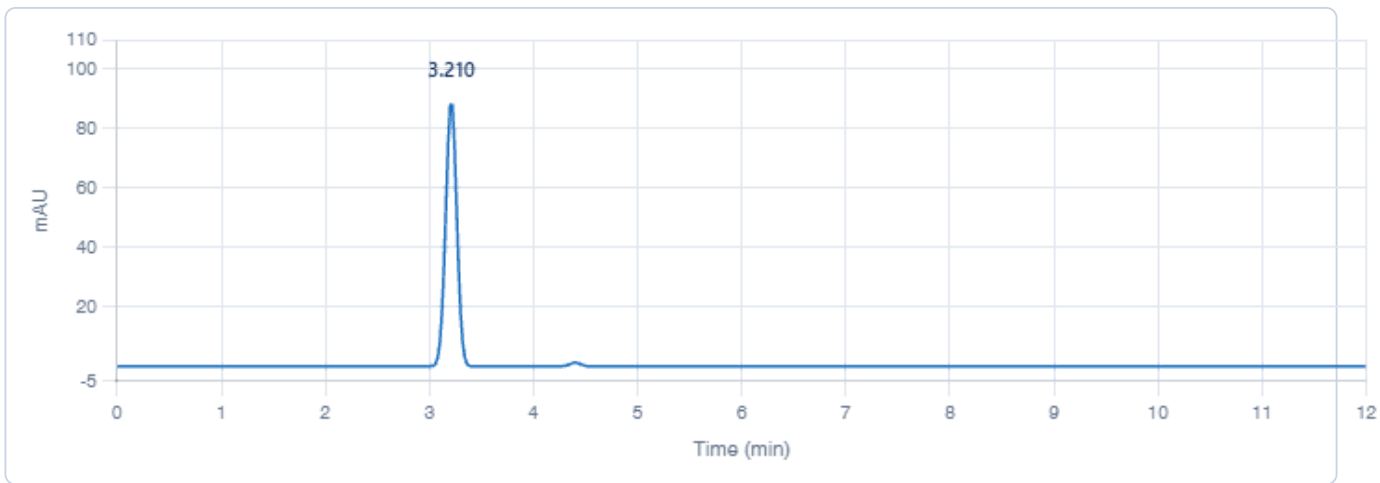
|                        |                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------|
| Product Name           | HGH 191aa 10 IU (Injectable Research Vial)                                                        |
| Chemical Name          | Somatropin (recombinant human growth hormone, 191 aa)                                             |
| CAS Number             | 12629-01-5                                                                                        |
| Batch / Lot Number     | HGHAA10IU260412                                                                                   |
| Manufacturing Date     | 2026-04-12                                                                                        |
| Retest / Expiry Date   | 2028-04-12                                                                                        |
| Net Content            | 10 IU per vial                                                                                    |
| Storage Conditions     | Store at 2–8°C, protected from light and moisture                                                 |
| Appearance             | White to off-white lyophilized powder                                                             |
| Molecular Formula / MW | C <sub>990</sub> H <sub>1528</sub> N <sub>262</sub> O <sub>300</sub> S <sub>7</sub> / See formula |

## ANALYTICAL RESULTS

| Test Item       | Specification                         | Result    | Method       |
|-----------------|---------------------------------------|-----------|--------------|
| Identification  | Conforms                              | Conforms  | LC-MS        |
| Purity (HPLC)   | ≥98.0%                                | 99.04%    | HPLC         |
| Peptide Content | 95.0–105.0%                           | 99.6%     | HPLC         |
| Appearance      | White to off-white lyophilized powder | Conforms  | Visual       |
| Water Content   | ≤3.0%                                 | 0.7%      | Karl Fischer |
| Heavy Metals    | ≤10 ppm                               | <10 ppm   | ICP-MS       |
| Endotoxin       | <10 EU/mg                             | <10 EU/mg | LAL Test     |

## HPLC CHROMATOGRAM

DAD1A, Sig=214, 4 Ref=off



*For laboratory research use only. Not for human consumption.*

TESTER NAME

Dr. Michael Carter

SIGNATURE

*Michael Carter*

TEST DATE

2026-04-30